Madhuvarsu Rupesh Chandra Bharadwaj

+91 8500617107 rupesh.2k5chandra@gmail.com GitHub LeetCode Portfolio

Profile

Final-year Computer Science student with hands-on experience in software development, backend services, REST APIs, databases, AI applications, testing, debugging, and secure application development. Experienced in building and deploying end-to-end applications and solving real-world engineering problems.

Education

VIT-AP University **2023 – Present**
B.Tech in Computer Science and Engineering **CGPA: 8.3**
Narayana Junior College, Nellore **2021 – 2023**
Intermediate **95%**

Technical Skills

Languages: Python, Java, JavaScript, SQL
Software Development: Data Structures, OOP, REST APIs, API Integration, JSON, Authentication, RBAC, Debugging, Unit Testing
Backend: Node.js, Express.js, PostgreSQL, MongoDB, Mongoose, Prisma
Frontend: TypeScript, React.js, HTML5, CSS3, Tailwind CSS
AI/ML: Machine Learning, Generative AI, LLMs, NLP, Prompt Engineering, Reinforcement Learning, PPO
Tools: Git, GitHub, Postman, Vercel, CI/CD

Experience

Backend Lead *STON Technology*
Software Development & Backend Engineering

- Developed and maintained backend services using Node.js, Express.js, and MongoDB for a multi-tenant college placement management platform.
- Implemented secure registration workflows, role-based access control (RBAC), tenant isolation, and department onboarding.
- Investigated and fixed backend issues, implemented secure document upload workflows, and added regression tests.

Full-Stack Developer – Client Project **Velora**

- Developed and deployed a production-oriented full-stack car rental platform supporting customer bookings and vehicle-owner management.
- Implemented JWT authentication, role-based access, MongoDB data management, API integration, debugging, and application deployment.

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, REST APIs, Razorpay, Git, Vercel

Projects

Smart Candidate Screening System Live Demo
Tech Stack: React.js, TypeScript, Node.js, Express.js, PostgreSQL, Prisma

- Built a full-stack platform to automate student screening and reduce manual shortlisting for college placement teams.
- Developed an explainable scoring system using CGPA, skills, projects, internships, and certifications to categorize candidates.

AI Bilingual Voice Assistant **GitHub**
Tech Stack: Python, LLM APIs, Speech-to-Text, Text-to-Speech, Prompt Engineering

- Built a bilingual AI application using speech-to-text, contextual LLM processing, and text-to-speech for conversational interaction.
- Designed structured prompts, validation logic, and contextual boundaries to improve response reliability.

Hybrid LLM-RL NPC Behaviour **GitHub**
Tech Stack: Python, PPO, Stable-Baselines3, Reinforcement Learning, LLMs

- Developed a Python-based RPG AI system combining PPO-based Reinforcement Learning with an LLM explanation layer.
- Designed a 78-dimensional observation space and four personality-based decision strategies for tactical NPC behaviour.
- Evaluated system performance using win rate and threat-targeting accuracy, achieving 75%+ win rate and 85% threat-targeting accuracy.